Volcano ECDLP census follow-up, Phase C: analysis and report

Study `research/volcano_ecdlp_census_followup_20261023/`; plan `PLAN.md`, analysis plan `ANALYSIS_PLAN.md` (frozen before the first Phase B session), amendments `AMENDMENTS.md` (A1, A2, and **A3, post hoc**, written after the data and before the final run). Every number below is read from `analysis/results.json` (written by `analysis/analyze.py`; 10,000 bootstrap or permutation replicates; seeds 20261022, 20261023, 20261024). **Status: observed, not established.** No speedup is claimed beyond H-T's measured ratio. Independent validation (`VALIDATION.json`) has not happened; this is the analysis, not its validation (independent validation round 1 failed this report on one material finding and six minor ones; see the Erratum below, which supersedes the text it names). It extends the validated census `research/volcano_ecdlp_census_20261017/`.

Erratum (round 1, post hoc)

Written after the independent validator's round-1 report (`benchmarks/20261023-volcano-census-followup/VALIDATOR_REPORT.md`, findings F1-F7). It corrects wording and states row sets; no number, seed, frozen rule or `results.json` value changed (`analyze.py` was not re-run; the record is `AMENDMENTS.md`, "Erratum (round 1, post hoc)"). Where the original text below conflicts with this section, this section governs.

- **F1 (material), H-M verdict.** With the frozen seed (20261024) and 10,000 shuffles, `ic` is significant on 5 of 12 curves, and the frozen rule reads that as inconclusive. The 5-versus-6 split rests on one curve, V6:98: F = 2.81, Holm p = 0.0511 (permutation p = 0.0073, Holm multiplier 7). Independent evaluations of the same test put that curve on the other side of 0.05: the validator's own 10,000-shuffle run with the same seed gives p 0.0064 and Holm 0.0448; a 2,000,000-shuffle run gives p 0.00711 (Monte Carlo SE 0.00006) and Holm 0.0498 (SE 0.0004); the parametric F(7, 504) test gives Holm 0.0489. All three give 6 of 12 for `ic`, with `rho-neg` 0 and `bsgs` 0, which the frozen rule reads as supported for IC. **The H-M verdict therefore sits on the Monte Carlo boundary between inconclusive and supported for IC and is not robust either way.** The frozen-seed result (5 of 12, inconclusive) is kept as the reported value, as are the section 8 table and figure 8. Two phrases are **withdrawn**: 'absent on others' and 'neither a general property nor noise'. A curve that is not significant after Holm is not shown to lack a model effect, and the statement about IC's model dependence is limited to this: `ic` is significant on 5 or 6 of 12 curves, `rho-neg` and `bsgs` on none.
- **F2, V6 level effect.** The heading of item 2 is read as: 19 of 20 V6 level ratios equivalent, 1 inconclusive (`rho-neg` level 2, 90% CI upper end 1.0535 against the margin 1.0526), none different. 'No method depends on level' is withdrawn as a summary of all 20 cells. Two cells lie within about 2 Monte Carlo standard errors of the TOST margin: `rho-neg` level 2 (above) and `rho-plain` level 1 (90% CI lower end 0.9508 against 0.95); the independent recomputation put both on the same side.
- **F3, H-T reference.** All 63 source vertices have j not in {0, 1728} on a prime field, so `rho-neg` is the strongest available rho at every source. H-T is relative to `rho-neg` at the source, which is the strongest rho available on every (j != 0) source curve. Item 4's last clause and the section 12 H-T bullet are corrected accordingly.
- **F4, wall admission row sets.** The A/A ratios (V5 1.042, V6 1.001) and the per-arm L2 fractions of section 10 use the isogenous families only (D, Ua, Ub, S, N). ANALYSIS_PLAN s8 also lists ctl; with ctl the validator finds A/A 1.0393 (V5, 22,384 pairs) and 1.0019 (V6, 44,352 pairs), still inside [0.95, 1.05], and the per-arm L2 fractions rise by 0.4-3.6 points and stay below 100%, so wall remains 'not admitted'. The 87% (V5) and 84% (V6) of item 7 and section 12 count all rows of the volcano (M and Tcr included); the section 10 values are per arm over isogenous rows.
- **F5, A2.1 time span.** The nine busy-host refusals are timestamped 05:32:32Z to 05:43:01Z in `runs/progress.jsonl`, not 05:41Z to 05:43Z. The refusal message itself was not preserved in any session log. No data is affected.
- **F6, per-curve flags (section 5).** With 648 curves and the pooled-|z*| reference, the smallest attainable Holm-adjusted p of the per-curve test is about 0.33 (observed 0.331 at z = 5.12, vertex 545), so 'no curve flagged' could not have come out otherwise and carries no information. The statement 'no V6 curve differs from its isomorphic re-scalings beyond the control spread' rests only on the aggregate F tests (p 0.15 / 0.67 / 0.42).
- **F7, V5 bsgs constant ratios (section 4).** R(L) = 0.988 (`bsgs`) and 0.982 (`bsgs-neg`) at every V5 level is a family-mix artifact, not a level effect: see the note after the section 4 table.

1. Verdicts

1. **Q2a, positive control (V5 j = 0 crater): PASS.** mean S(`rho-cubic`) / mean S(`rho-neg`) = **0.6113**, 95% CI [0.5995, 0.6237] over 4000 paired runs on 3 Weierstrass models (theory 1/sqrt(3) = 0.5774; pass needs the upper bound below 0.85; slug-cluster sensitivity [0.5967, 0.6288]). The detector sees a cost reduction, so the 'equivalent' verdicts below are informative.
2. **Q1, V6 (l = 3, depth 4, 648 curves) level effect: 19 of 20 ratios equivalent, 1 inconclusive, none different** (erratum F2; the earlier heading 'no method depends on level' is withdrawn). Of the 20 V6 level ratios, 19 are equivalent (TOST, +-5% rho and BSGS, +-10% IC), 1 inconclusive (V6 `rho-neg` level 2: ratio 1.022, 90% CI not inside the margin; not different after Holm), none different. The census finding (no level effect) replicates on a deeper l = 3 volcano.
3. **Q2b, V5 (j = 0 crater, depth 6, 64 curves) level effect:** levels 4-6 (8, 16, 32 curves), 15 ratios: all 15 equivalent. Levels 1-3 (1, 2, 4 curves) are **descriptive** (ratios within −0.031 to +0.011 of 1). `rho-cubic` is excluded from the level analysis.
4. **H-T (carry up to the crater, solve with `rho-cubic`): SUPPORTED.** Pooled over 63 source vertices (vertex-weighted, A3.2), T = 0.5946 [0.5684, 0.6220] at 0.0 GAE, 0.5917 [0.5655, 0.6193] at 0.1 GAE, 0.5781 [0.5512, 0.6063] at 1.0 GAE per Velu point map. The control T0 (same transport, same crater curve, `rho-neg`) = 0.9672 [0.9227, 1.0145], 0.9672 [0.9223, 1.0149], 0.9670 [0.9204, 1.0164], whose CI includes 1 at all three prices. Theory 1/sqrt(3) = 0.5774; the transport term is 0.21% of the numerator at 1 GAE. Measured same-curve automorphism gain in S (`rho-cubic` / `rho-neg` on the carried instances): 0.6148 [0.5895, 0.6420]. This is a measured end-to-end ratio of a known composition, about 0.58x in ops_gae, not a new speedup; and it is relative to `rho-neg` at the source, which is the strongest rho available on every (j != 0) source curve (erratum F3).
5. **H-M (does IC cost depend on the Weierstrass model?): INCONCLUSIVE by the frozen-seed count, on the Monte Carlo boundary with 'supported for IC' (erratum F1).** `ic` is significant (Holm, alpha 0.05) on 5 of 12 chosen curves; `rho-neg` on 0 and

`bsgs` on 0. Supported would need 6 of 12 for `ic` and at most 1 for each of the others; refuted would need at most 1 for `ic`. [Withdrawn in the erratum (F1): the sentence 'IC's model dependence is therefore present on some curves and absent on others: it is neither a general property nor noise' is not supported; a precise evaluation of the same test gives 6 of 12, 'supported for IC', so the verdict is not robust either way.] The census's V3 crater values (IC 57 / 67 / 73) are quoted only as the motivating observation.

6. **IC against matched rho (distance remaining, not pass/fail).** V6: IC costs 60.4x `rho-neg` in S with nothing priced and 90.8x with every uncharged operation priced at 1 GAE. V5 levels 1-6: 53.3x and 81.8x. **On the V5 j = 0 crater the matched rho is `rho-cubic`: 89.3x / 98.2x / 141.2x at 0 / 0.1 / 1 GAE** (not the `rho-neg` ratio). The order (rho ahead) holds across the whole band at every level. No level improves IC's position.
7. **Wall time: not admitted** on either volcano. The A/A ratio is within [0.95, 1.05] (V5 1.042, V6 1.001), but only 87% (V5) and 84% (V6) of rows (all rows of the volcano, M and Tcr included; erratum F4; A/A ratios use isogenous rows, with ctl 1.0393 and 1.0019) reached isolation L2, so condition 1 fails. Wall tables are descriptive.

2. What was measured

- 147 audited ecbench sessions (V5: 7 D, 6 Ua, 6 Ub, 7 N, 7 ctl, 63 Tcr, 4 M; V6: 8 M and D, Ua, Ub, S, N, ctl), each `audit OK` with exact replays; 428,976 measured runs after dropping 241,712 warm-up runs. Unverified measured runs: 0.
- Arms: `rho-neg` (matched reference), `rho-plain`, `bsgs`, `bsgs-neg`, `ic` (`ic.pipeline`, `prime-abscissa:size=64`), `rho-neg-aa` (A/A), `rho-cubic` (V5 crater D, N, ctl and every Tcr spec). Tcr specs carry `rho-cubic` and `rho-neg`; M specs `rho-neg`, `bsgs`, `ic`.
- Populations: V5 (l = 2, p = 66678169, 26-bit, depth 6; 1/1/2/4/8/16/32 curves); V6 (l = 3, p = 38002051, depth 4; 8/16/48/144/432 curves). Curve = census vertex for every resample (A2.3 in force from the start).
- Unit S (normalised group operations); uncharged counters (`canonicalisations_uncharged` etc.) priced at 0, 0.1 and 1 GAE per operation from each session's `records.jsonl` (A2.1 in force). Totals agree with the CSV extracts row by row (max |dS| = 0.0, max |d ops_gae| = 0.0).
- Host: shared 4-vCPU cloud VM; frozen binary `ac4bc5b2...`; session wall 8.94 h, 7 recorded pauses (driver and container restarts) excluded from the wall budget.

3. Q2a positive control

V5 crater (vertex 0, j = 0; D, N and ctl rows, 3 models): ratio 0.6113, CI [0.5995, 0.6237]; theory 0.5774; census V3 value for comparison (not pooled) in the census report. **Pass.** `rho-cubic` is valid only at j = 0, which on V5 is the crater alone.

4. Q1 and Q2b: level effect

R(L) = mean S at level L / mean S at the crater, isogenous families pooled (D, Ua, Ub, N; plus S on V6), cluster bootstrap over census vertices stratified by level. Verdicts in the plan's vocabulary; Holm over 35 tests (A3.5).

V	arm	L	curves	R(L) on S	95% CI	90% CI	half-width	margin	Holm p	verdict
V6	rho-neg	1	16	1.016	[0.979, 1.056]	[0.985, 1.050]	0.039	0.95-1.05	1.000	equivalent
V6	rho-neg	2	48	1.022	[0.985, 1.059]	[0.991, 1.054]	0.037	0.95-1.05	1.000	inconclusive
V6	rho-neg	3	144	1.011	[0.978, 1.045]	[0.983, 1.040]	0.034	0.95-1.05	1.000	equivalent
V6	rho-neg	4	432	1.016	[0.986, 1.047]	[0.990, 1.042]	0.031	0.95-1.05	1.000	equivalent
V6	rho-plain	1	16	0.978	[0.945, 1.011]	[0.951, 1.005]	0.033	0.95-1.05	1.000	equivalent
V6	rho-plain	2	48	1.001	[0.968, 1.034]	[0.973, 1.029]	0.033	0.95-1.05	1.000	equivalent
V6	rho-plain	3	144	0.994	[0.963, 1.025]	[0.968, 1.020]	0.031	0.95-1.05	1.000	equivalent
V6	rho-plain	4	432	0.997	[0.968, 1.025]	[0.973, 1.021]	0.028	0.95-1.05	1.000	equivalent
V6	bsgs	1	16	1.000	[0.990, 1.010]	[0.992, 1.008]	0.010	0.95-1.05	1.000	equivalent
V6	bsgs	2	48	1.000	[0.991, 1.009]	[0.992, 1.008]	0.009	0.95-1.05	1.000	equivalent
V6	bsgs	3	144	1.000	[0.991, 1.009]	[0.993, 1.007]	0.009	0.95-1.05	1.000	equivalent
V6	bsgs	4	432	1.000	[0.992, 1.008]	[0.994, 1.007]	0.008	0.95-1.05	1.000	equivalent
V6	bsgs-neg	1	16	1.000	[0.986, 1.015]	[0.988, 1.012]	0.015	0.95-1.05	1.000	equivalent
V6	bsgs-neg	2	48	1.000	[0.987, 1.014]	[0.989, 1.012]	0.014	0.95-1.05	1.000	equivalent
V6	bsgs-neg	3	144	1.000	[0.987, 1.013]	[0.990, 1.011]	0.013	0.95-1.05	1.000	equivalent
V6	bsgs-neg	4	432	1.000	[0.989, 1.012]	[0.991, 1.010]	0.012	0.95-1.05	1.000	equivalent
V6	ic	1	16	0.969	[0.923, 1.018]	[0.929, 1.010]	0.048	0.90-1.11	1.000	equivalent
V6	ic	2	48	0.972	[0.934, 1.015]	[0.939, 1.008]	0.041	0.90-1.11	1.000	equivalent
V6	ic	3	144	0.972	[0.935, 1.011]	[0.941, 1.005]	0.038	0.90-1.11	1.000	equivalent
V6	ic	4	432	0.967	[0.933, 1.004]	[0.939, 0.998]	0.035	0.90-1.11	1.000	equivalent
V5	rho-neg	1	1	1.011	[0.983, 1.041]	[0.987, 1.036]	0.029	0.95-1.05	-	descriptive

V5	rho-neg	2	2	0.988	[0.961, 1.016]	[0.965, 1.012]	0.027	0.95-1.05	-	descriptive
V5	rho-neg	3	4	1.006	[0.973, 1.041]	[0.978, 1.035]	0.034	0.95-1.05	-	descriptive
V5	rho-neg	4	8	1.001	[0.962, 1.037]	[0.969, 1.032]	0.037	0.95-1.05	1.000	equivalent
V5	rho-neg	5	16	1.010	[0.979, 1.043]	[0.983, 1.038]	0.032	0.95-1.05	1.000	equivalent
V5	rho-neg	6	32	1.018	[0.985, 1.053]	[0.990, 1.047]	0.034	0.95-1.05	1.000	equivalent
V5	rho-plain	1	1	0.969	[0.941, 0.998]	[0.945, 0.993]	0.028	0.95-1.05	-	descriptive
V5	rho-plain	2	2	0.998	[0.969, 1.029]	[0.973, 1.024]	0.030	0.95-1.05	-	descriptive
V5	rho-plain	3	4	0.973	[0.944, 1.005]	[0.948, 1.000]	0.031	0.95-1.05	-	descriptive
V5	rho-plain	4	8	0.991	[0.957, 1.027]	[0.962, 1.021]	0.035	0.95-1.05	1.000	equivalent
V5	rho-plain	5	16	0.995	[0.964, 1.027]	[0.969, 1.022]	0.032	0.95-1.05	1.000	equivalent
V5	rho-plain	6	32	0.987	[0.949, 1.027]	[0.955, 1.020]	0.039	0.95-1.05	1.000	equivalent
V5	bsgs	1	1	0.988	[0.974, 1.002]	[0.976, 1.000]	0.014	0.95-1.05	-	descriptive
V5	bsgs	2	2	0.988	[0.974, 1.002]	[0.976, 1.000]	0.014	0.95-1.05	-	descriptive
V5	bsgs	3	4	0.988	[0.974, 1.002]	[0.976, 1.000]	0.014	0.95-1.05	-	descriptive
V5	bsgs	4	8	0.988	[0.974, 1.002]	[0.976, 1.000]	0.014	0.95-1.05	1.000	equivalent
V5	bsgs	5	16	0.988	[0.975, 1.002]	[0.977, 1.000]	0.014	0.95-1.05	1.000	equivalent
V5	bsgs	6	32	0.988	[0.974, 1.002]	[0.976, 1.000]	0.014	0.95-1.05	1.000	equivalent
V5	bsgs-neg	1	1	0.982	[0.961, 1.004]	[0.964, 1.000]	0.022	0.95-1.05	-	descriptive
V5	bsgs-neg	2	2	0.982	[0.961, 1.003]	[0.964, 1.000]	0.021	0.95-1.05	-	descriptive
V5	bsgs-neg	3	4	0.982	[0.961, 1.003]	[0.964, 1.000]	0.021	0.95-1.05	-	descriptive
V5	bsgs-neg	4	8	0.982	[0.960, 1.003]	[0.964, 1.000]	0.021	0.95-1.05	1.000	equivalent
V5	bsgs-neg	5	16	0.982	[0.961, 1.004]	[0.964, 1.000]	0.021	0.95-1.05	1.000	equivalent
V5	bsgs-neg	6	32	0.982	[0.960, 1.003]	[0.964, 1.000]	0.022	0.95-1.05	1.000	equivalent
V5	ic	1	1	0.973	[0.952, 0.995]	[0.955, 0.991]	0.021	0.90-1.11	-	descriptive
V5	ic	2	2	0.970	[0.940, 1.001]	[0.945, 0.996]	0.030	0.90-1.11	-	descriptive
V5	ic	3	4	0.986	[0.946, 1.025]	[0.953, 1.019]	0.039	0.90-1.11	-	descriptive
V5	ic	4	8	0.985	[0.956, 1.013]	[0.960, 1.009]	0.029	0.90-1.11	1.000	equivalent
V5	ic	5	16	0.987	[0.959, 1.017]	[0.963, 1.012]	0.029	0.90-1.11	1.000	equivalent
V5	ic	6	32	1.011	[0.983, 1.040]	[0.987, 1.035]	0.029	0.90-1.11	1.000	equivalent

Constant V5 BSGS ratios (erratum F7). V5 `bsgs` 0.988 and `bsgs-neg` 0.982 at every level 1-6 are a family-mix artifact, not a level effect: Ua and Ub have no V5 level-0 rows, so the crater mean is D + N while each other level is D + Ua + Ub + N. BSGS cost depends only on the instance's k; D, Ua and Ub share a target seed and N has its own, with mean S(N)/S(D) = 1.0484 (`bsgs`) and 1.0753 (`bsgs-neg`). With x that ratio, (3 + x) / (2 (1 + x)) = 0.98819 and 0.98186 reproduce the two constants. The same mix difference enters every V5 arm's R(L), negligibly for rho and ic (the census validation flagged the same pattern, its F8).

Secondary mixed models (`log S ~ C(level) + C(family) + (1 | vertex)`, REML) converge for `rho-neg`, `rho-plain` and `ic` on both volcanoes and for V6 `bsgs`; they do not converge for V5 `bsgs`, V5 `bsgs-neg` and V6 `bsgs-neg` (BSGS cost is a deterministic function of r and k, so the curve variance is degenerate; reported as not converged in `results.json`). ICC V6 `ic` 0.0050, V5 `ic` 0.0035: curves differ very little beyond their rows. They never override the bootstrap. Per-family R(L) (D, Ua, Ub, N, S) is in `results.json` (descriptive; Ua and Ub have no V5 level-0 rows, A3.6). See figures 1, 2 and 6.

5. Curve effect against the isomorphic negative control (V6)

arm	F	permutation p	Holm p	curves flagged (adj. p < 0.01)
rho-neg	1.04	0.149	0.448	0
rho-plain	0.98	0.674	0.849	0
bsgs	n/a (zero within-vertex variance)	-	-	0
bsgs-neg	n/a (zero within-vertex variance)	-	-	0
ic	0.99	0.425	0.849	0

No variance-ratio test is significant and no curve is flagged: no V6 curve differs from its own isomorphic re-scalings beyond the control spread (figure 4). **Erratum F6:** the per-curve flag cannot fire at this size (smallest attainable Holm-adjusted p about 0.33; observed 0.331), so the sentence rests only on the aggregate F tests (p 0.15 / 0.67 / 0.42) and 'no curve is flagged' is uninformative.

6. Transfer (V6)

- **Recovery:** every one of the 428,976 measured runs on both volcanoes was verified (smallest Clopper-Pearson 95% lower bound 0.9963, for the smallest cell).
- BSGS counts are identical across D, Ua, Ub and S in 100% of triples on both volcanoes.
- **Paired source ratios** (Ua, Ub, S, N over D; `rho-neg`, `rho-plain`, `ic`; Holm family of 12): 12 of 12 equivalent, 0 inconclusive, 0 different.
- **Transport cost:** about 3172 ns per edge on V6 and 3669 ns on V5, 2 Velu point maps per edge; the largest transport/solve wall fraction is 1.2% (V5) and 2.6% (V6) of a solve (figure 5; wall is descriptive).
- **Transport, then solve (V6):** 2 of 100 (family, arm, destination level) cells meet the plan's literal 'signal' rule: `V6|S|ic` destination level 1, `V6|S|ic` destination level 3. Both are `ic` on the family S, whose source is one vertex (the crater-b vertex); the same family and arm have the same point ratio (0.85-0.89) at the other three levels, whose CIs reach 1 or overlap the control's, and the family D and Ua `ic` rows are at 0.88-0.92 and 0.76-0.81 without meeting the rule. The source mean comes from a single curve (the plan warns it is noisy and shared by all destination levels), the native control ratio is itself above 1, and no `rho-neg` or `rho-plain` cell meets the rule. It is labelled a signal under the rule and **is not a speedup claim**: IC remains 50x or more behind `rho-neg` at every level, so a 10 to 15% shift in IC's own cost moves nothing about the distance to rho.

7. H-T: carry up, then solve with the automorphism (V5)

For each of the 63 non-crater vertices v the instance born at v is carried to the crater (verified isogenies; the last edge outside `ca` by the study's helper, checked), and solved there with `rho-cubic`. T_v(p) = [mean priced ops_gae(`rho-cubic`, Tcr_v) + p * M_v] / mean priced ops_gae(`rho-neg`, N_v); T0 is the same with `rho-neg` on Tcr_v. M_v = Velu point maps of the path + 2 for the last edge (2 to 12). Checks: unpaired rows 0, path-length check failures 0.

stratum	vertices	T at 0 GAE	T at 0.1	T at 1	T0 at 0	T0 at 1
overall (verdict)	63	0.5946 [0.5684, 0.6220]	0.5917 [0.5655, 0.6193]	0.5781 [0.5512, 0.6063]	0.9672 [0.9227, 1.0145]	0.9670 [0.9204, 1.0164]
level 1 (descriptive)	1	0.599 [0.567, 0.633]	0.596 [0.564, 0.630]	0.582 [0.549, 0.617]	0.958 [0.905, 1.013]	0.956 [0.902, 1.014]
level 2 (descriptive)	2	0.600 [0.565, 0.637]	0.597 [0.562, 0.634]	0.583 [0.546, 0.621]	1.005 [0.945, 1.069]	1.005 [0.943, 1.073]
level 3 (descriptive)	4	0.581 [0.547, 0.618]	0.578 [0.543, 0.615]	0.563 [0.528, 0.601]	1.013 [0.952, 1.080]	1.015 [0.951, 1.085]
level 4 (descriptive)	8	0.628 [0.577, 0.688]	0.625 [0.574, 0.685]	0.612 [0.559, 0.674]	1.006 [0.938, 1.081]	1.007 [0.936, 1.086]
level 5 (descriptive)	16	0.615 [0.567, 0.667]	0.612 [0.564, 0.664]	0.599 [0.550, 0.653]	1.006 [0.917, 1.098]	1.007 [0.915, 1.104]
level 6 (descriptive)	32	0.578 [0.536, 0.623]	0.575 [0.533, 0.621]	0.562 [0.518, 0.608]	0.932 [0.859, 1.008]	0.931 [0.855, 1.010]

Verdict rule (plan s4, applied at each price): supported if the overall CI of T is below 1 at all three prices and the CI of T0 includes 1 at all three; refuted if T's CI includes 1 or lies above it at any price. Inputs: T below 1 at all prices = True, T0 includes 1 at all prices = True, T includes or exceeds 1 at some price = False. **Verdict: supported.** Per-vertex T_v at 1 GAE ranges 0.43 to 0.89 (pair-bootstrap CIs, figure 7); per-level values are descriptive. Level 6 has T0 = 0.931 (the control itself sits below 1 there, descriptive), and level 1 0.956; the pooled T0 interval includes 1. Per the census lesson F2, a per-curve difference is read against isomorphic controls: T0 is that control, and it shows the spread the carried instance alone has.

8. H-M: does IC cost depend on the Weierstrass model?

Eight isomorphic models (u not +-1) per chosen curve, native generator carried by the isomorphism, 16 targets x 4 rounds = 64 solves per model and arm. F = between-model / within-model mean square; permutation p (10,000 shuffles of rows across models, seed 20261024); Holm over the 12 curves of each arm. BSGS must be model independent; it is (F = 0 on every curve; identical ops_gae across models for each (target, round) in 100% of cases), so the models carry the same instances.

curve	level	`ic` F	`ic` Holm p	`ic` max/min model mean (95% CI)	`rho-neg` F	`rho-neg` Holm p	`rho-neg` max/min (95% CI)
V5:0	0	1.95	0.2516	1.21 [1.15, 1.38]	1.09	1.000	1.19 [1.14, 1.42]
V5:14	4	3.21	0.0140 *	1.22 [1.17, 1.41]	0.70	1.000	1.14 [1.12, 1.39]
V5:19	5	1.77	0.2787	1.20 [1.14, 1.37]	0.38	1.000	1.12 [1.10, 1.36]
V5:44	6	0.77	0.8623	1.12 [1.09, 1.30]	0.99	1.000	1.19 [1.13, 1.44]
V6:2	0	3.22	0.0225 *	1.31 [1.24, 1.55]	0.40	1.000	1.10 [1.11, 1.38]
V6:214	1	5.91	0.0012 *	1.58 [1.39, 1.84]	0.62	1.000	1.15 [1.11, 1.37]
V6:223	3	3.21	0.0225 *	1.32 [1.23, 1.55]	1.09	1.000	1.20 [1.14, 1.44]
V6:262	2	1.00	0.8623	1.17 [1.13, 1.40]	0.74	1.000	1.15 [1.12, 1.39]
V6:43	2	2.21	0.1610	1.32 [1.20, 1.55]	0.48	1.000	1.13 [1.11, 1.37]

V6:599	4	2.33	0.1452	1.30 [1.17, 1.53]	1.18	1.000	1.15 [1.13, 1.41]
V6:98	0	2.81	0.0511	1.36 [1.23, 1.62]	1.82	0.991	1.26 [1.17, 1.50]
V6:99	1	5.23	0.0012 *	1.32 [1.28, 1.59]	0.71	1.000	1.14 [1.11, 1.37]

* = significant after Holm. Counts: `ic` 5 of 12, `rho-neg` 0, `bsgs` 0. **Verdict: inconclusive.** The max/min of eight model means of a quantity with sampling noise is above 1 by chance (rho-neg shows 1.10-1.26 with no significant curve); IC's values reach 1.58 on one curve. Two related observations from this study, descriptive only: the V5 crater's own D/N model and its two ctl re-scalings give IC mean S 59.3, 59.2, 58.1 (no model gap on that curve, unlike the census V3 crater), and the V6 curve-effect test (section 5) finds no curve differing from its re-scalings. The test permutes rows across models and ignores the shared (target, round) pairing, which makes it conservative (A3.7). With 5 of 12 the verdict is inconclusive by the plan's rule, which needed 6. **Erratum F1:** the sixth curve, V6:98 (Holm 0.0511), is within Monte Carlo error of 0.05; the validator's independent 10,000-shuffle run (Holm 0.0448), a 2,000,000-shuffle run (Holm 0.0498 +- 0.0004) and the parametric F test (Holm 0.0489) all give 6 of 12, 'supported for IC'. The verdict is on the boundary between inconclusive and supported for IC and is not robust either way; the frozen-seed count is reported as computed. The max/min CIs in the table are percentile bootstrap intervals of a maximum, biased upward, and cannot be read against 1: read IC's max/min against `rho-neg`'s on the same curves (validator note N1). 'Absent on others' and 'neither a general property nor noise' are withdrawn.

9. IC against matched rho (distance remaining)

Mean S(`ic`) / mean S(`rho-neg`), pairs of the same (workload, target, round), uncharged counts priced at 0 / 0.1 / 1 GAE on both arms (an arm with an unemitted count is a lower bound). Ranked only if the order holds at all three prices. V5 level 0 uses `rho-cubic` (the strongest rho at j = 0).

V	level	matched rho	IC/rho at 0 GAE	at 0.1	at 1	order across the band
V5	0	**rho-cubic**	89.3 [87.1, 91.6]	98.2 [95.8, 100.7]	141.2 [137.6, 144.9]	rho-cubic ahead of ic at all three prices
V5	1	rho-neg	52.3 [51.1, 53.7]	57.2 [55.8, 58.7]	80.3 [78.2, 82.4]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V5	2	rho-neg	53.4 [51.7, 55.1]	58.4 [56.5, 60.3]	81.9 [79.3, 84.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V5	3	rho-neg	53.3 [51.3, 55.3]	58.3 [56.1, 60.5]	81.8 [78.7, 84.9]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V5	4	rho-neg	53.5 [51.2, 56.0]	58.5 [56.0, 61.3]	82.1 [78.5, 86.1]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V5	5	rho-neg	53.1 [51.5, 54.8]	58.1 [56.3, 60.0]	81.5 [79.0, 84.2]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V5	6	rho-neg	54.0 [52.2, 55.9]	59.1 [57.1, 61.2]	82.9 [80.0, 85.9]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V6	0	rho-neg	63.1 [60.9, 65.3]	68.6 [66.2, 71.0]	95.1 [91.6, 98.5]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V6	1	rho-neg	60.1 [57.3, 62.8]	65.3 [62.2, 68.2]	90.4 [85.9, 94.6]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V6	2	rho-neg	60.1 [58.2, 62.0]	65.2 [63.2, 67.3]	90.2 [87.3, 93.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V6	3	rho-neg	60.6 [59.2, 62.1]	65.9 [64.3, 67.5]	91.2 [88.9, 93.5]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
V6	4	rho-neg	60.1 [59.3, 60.9]	65.3 [64.4, 66.2]	90.3 [89.0, 91.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)

pooled	at 0 GAE	at 0.1	at 1
V5 (levels 1-6, rho-neg)	53.3 [52.5, 54.0]	58.3 [57.5, 59.1]	81.8 [80.6, 83.0]
V6 (levels 0-4, rho-neg)	60.4 [59.8, 61.1]	65.6 [64.9, 66.3]	90.8 [89.8, 91.8]

The 12 M curves give per-curve IC/`rho-neg` of 51.8 to 65.2 at 0 GAE and 79.5 to 98.4 at 1 GAE (descriptive; on the V5 crater curve that is not the matched rho). Pricing the uncharged counters raises IC's cost more than rho's, so the gap widens with the price; it never closes at any of the three prices, and no level of either volcano moves it. Figure 3.

10. Wall time

V	A/A ratio	95% CI	A/A within 5%	rows at L2 per arm	admitted
V5	1.0424	[1.0376, 1.0472]	True	bsgs 86%, bsgs-neg 87%, ic 88%, rho-neg 86%, rho-plain 86%	False
V6	1.0005	[0.9969, 1.0040]	True	bsgs 77%, bsgs-neg 83%, ic 88%, rho-neg 79%, rho-plain 79%	False

M and Tcr specs have no A/A arm: their wall times are descriptive. Row sets (erratum F4): the table uses the isogenous families only; ANALYSIS_PLAN s8 also lists ctl, with which the A/A ratio is 1.0393 (V5) and 1.0019 (V6) and the per-arm L2 fractions are 0.4-3.6 points higher, still below 100%; 'not admitted' holds either way.

11. Figures

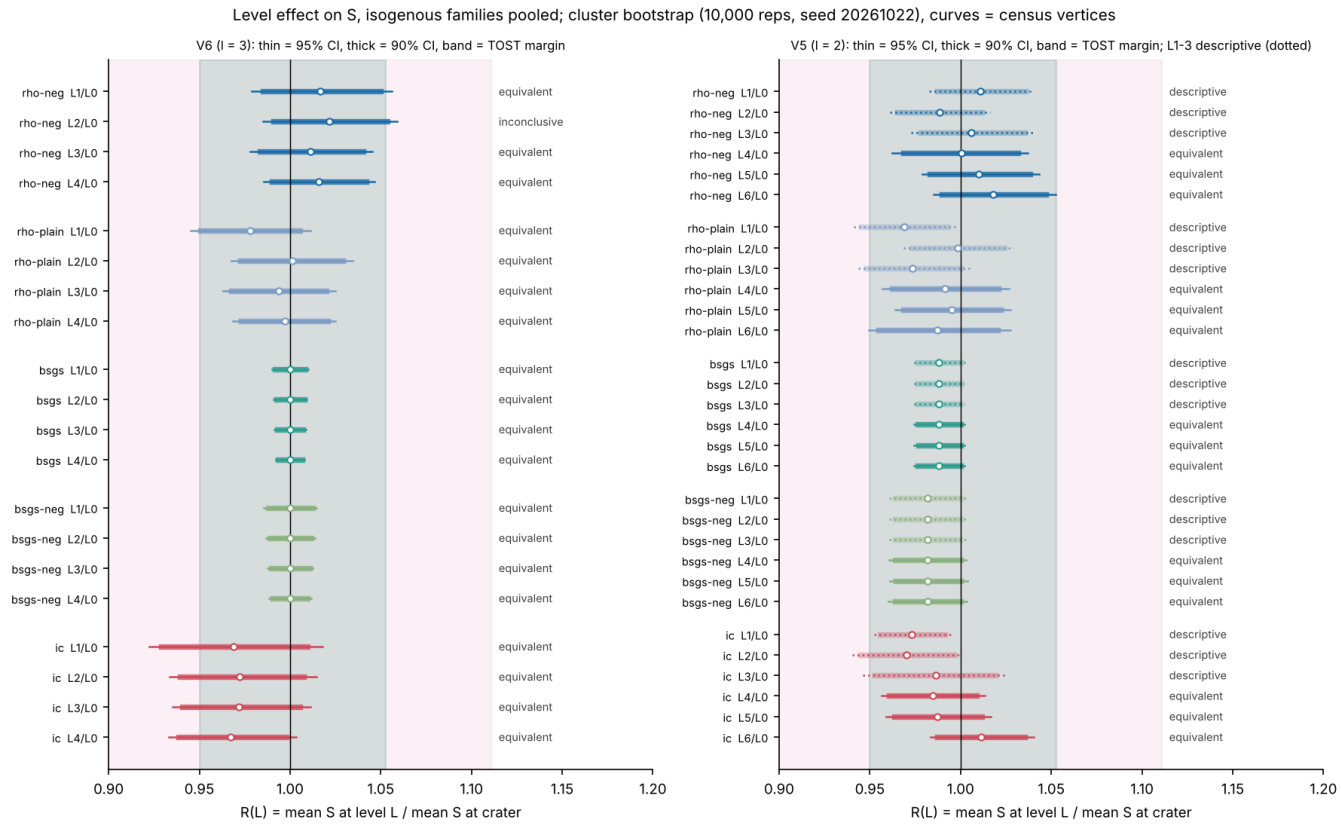


Figure 1: R(L) per volcano and arm with 90% and 95% CIs against the TOST band (V5 levels 1-3 dotted: descriptive)

fig1_forest_level_effect: R(L) per volcano and arm with 90% and 95% CIs against the TOST band (V5 levels 1-3 dotted: descriptive). Source `analysis/results.json`*; SVG beside the PNG.*

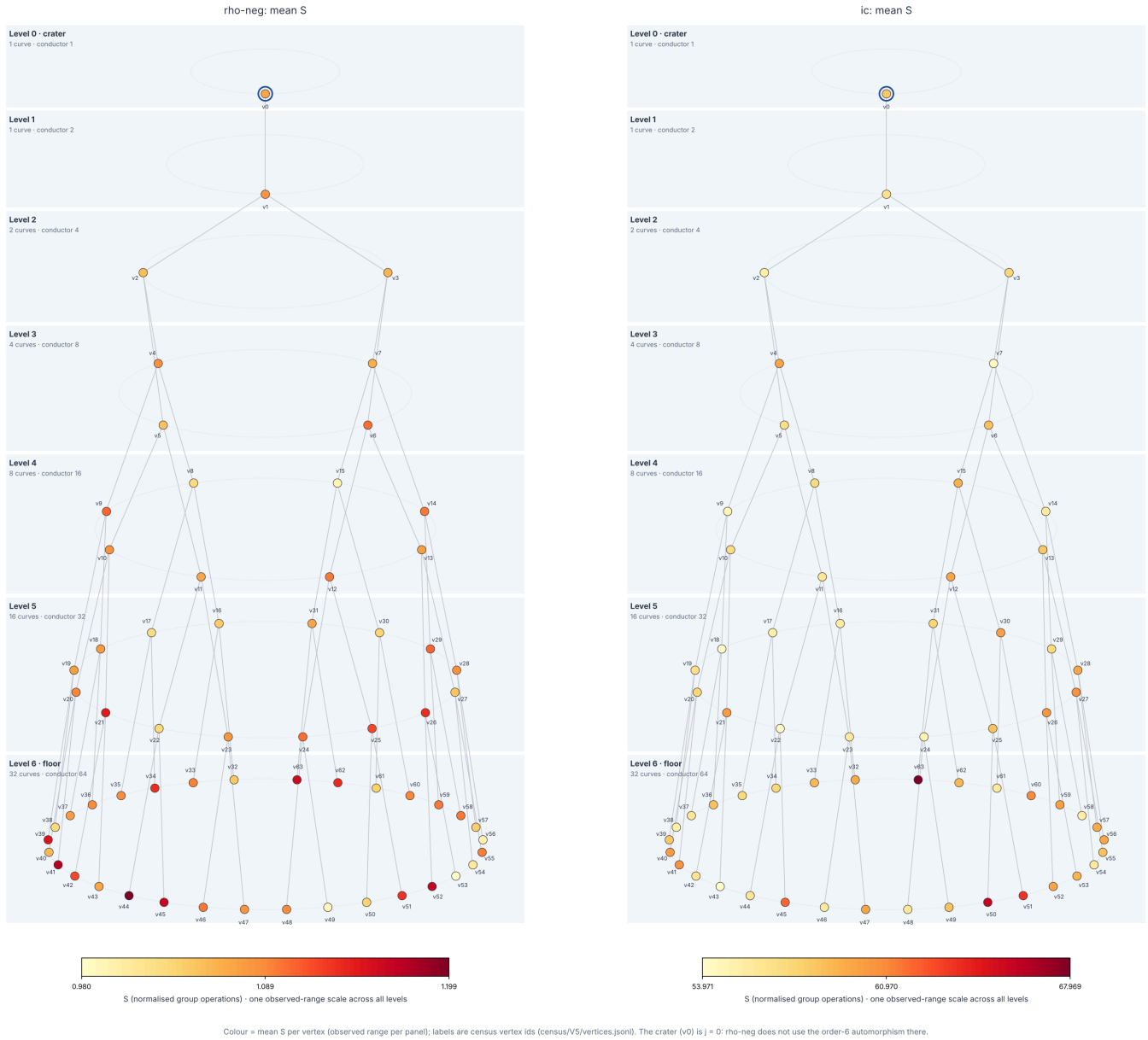


Figure 2: V5 volcano, mean S per curve, rho-neg and ic (standard `tools/volcano-diagram`)

fig2_volcano_V5: V5 volcano, mean S per curve, rho-neg and ic (standard `tools/volcano-diagram`). Source `analysis/results.json`; SVG beside the PNG.

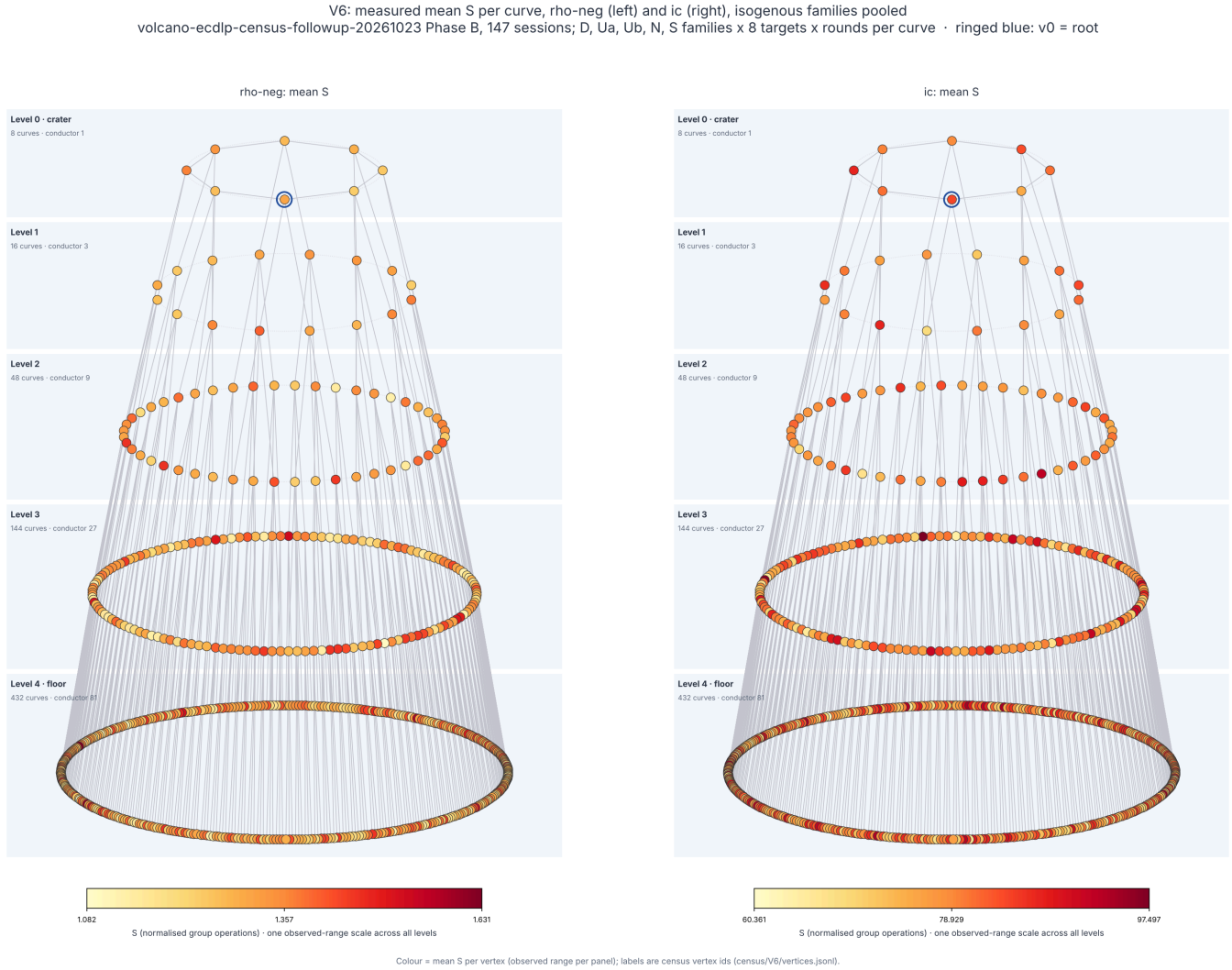


Figure 3: V6 volcano, same

fig2_volcano_V6: V6 volcano, same. Source `analysis/results.json`; SVG beside the PNG.

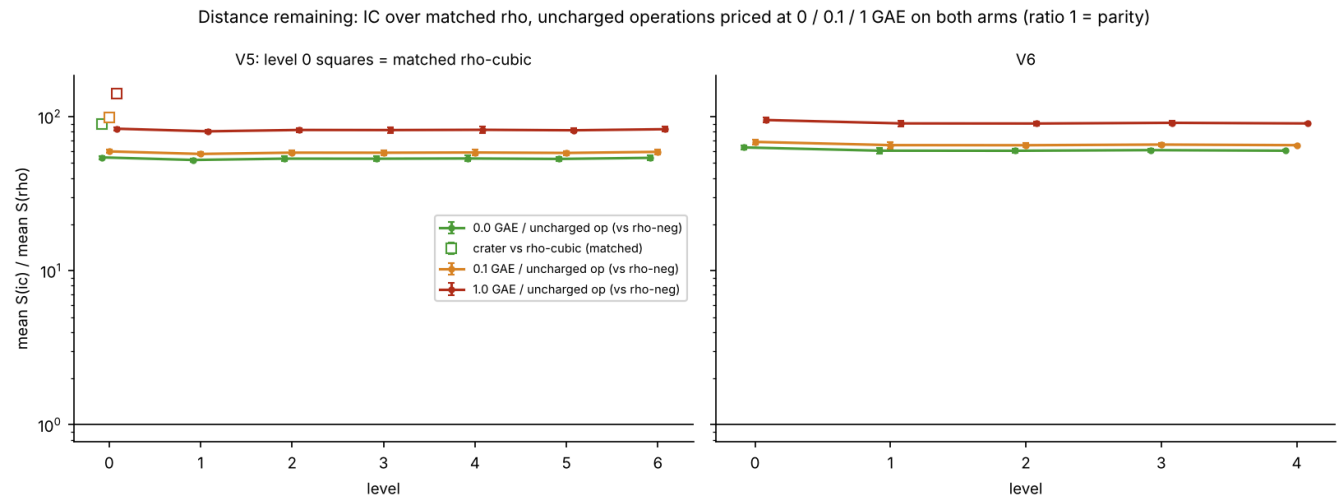


Figure 4: IC / matched rho by level at 0, 0.1, 1 GAE; V5 level 0 squares are rho-cubic

fig3_ic_over_rho: IC / matched rho by level at 0, 0.1, 1 GAE; V5 level 0 squares are rho-cubic. Source `analysis/results.json`; SVG beside the PNG.

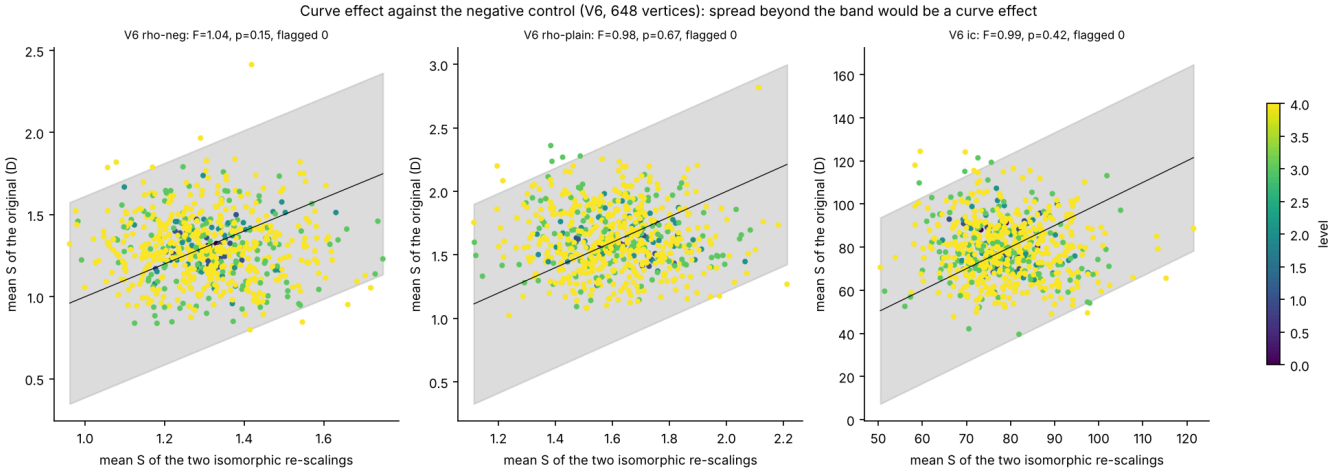


Figure 5: V6: each curve's mean S against its two isomorphic re-scalings

fig4_curve_vs_control: V6: each curve's mean S against its two isomorphic re-scalings. Source `analysis/results.json`; SVG beside the PNG.

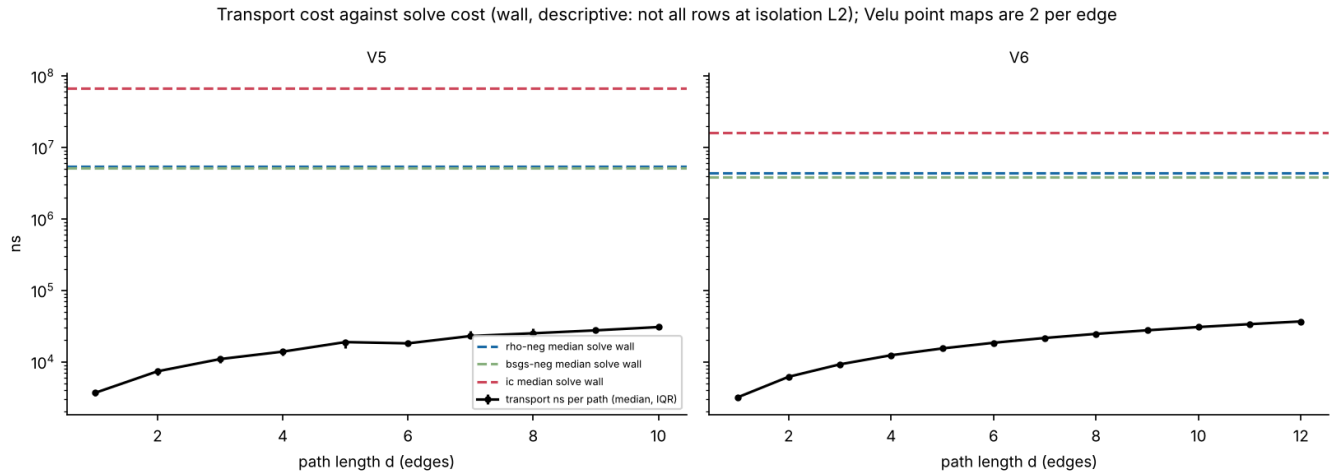


Figure 6: transport ns per path against median solve wall, by path length

fig5_transport_vs_solve: transport ns per path against median solve wall, by path length. Source `analysis/results.json`; SVG beside the PNG.

CV and achieved precision (mean over levels); BSGS is deterministic in r and k, so its CV is the scalar's spread only

volcano	arm	curves per level	rows	CV of S per row	95% half-width of level mean	95% half-width of R(L)	ICC (mixed model)	converged
V5	rho-neg	1/1/2/4/8/16/32	14256	0.46	2.4%	0.032	0.000	yes
V5	rho-plain	1/1/2/4/8/16/32	14256	0.48	2.5%	0.033	0.000	yes
V5	bsgs	1/1/2/4/8/16/32	14256	0.23	1.0%	0.014	n/a	no
V5	bsgs-neg	1/1/2/4/8/16/32	14256	0.36	1.6%	0.021	n/a	no
V5	ic	1/1/2/4/8/16/32	14256	0.36	2.4%	0.029	0.003	yes
V6	rho-neg	8/16/48/144/432	31680	0.45	2.0%	0.035	0.001	yes
V6	rho-plain	8/16/48/144/432	31680	0.45	1.9%	0.031	0.000	yes
V6	bsgs	8/16/48/144/432	31680	0.17	0.5%	0.009	0.000	yes
V6	bsgs-neg	8/16/48/144/432	31680	0.26	0.8%	0.013	0.000	no
V6	ic	8/16/48/144/432	31680	0.46	2.4%	0.040	0.005	yes

Figure 7: CV and achieved precision

fig6_cv_precision: CV and achieved precision. Source `analysis/results.json`; SVG beside the PNG.

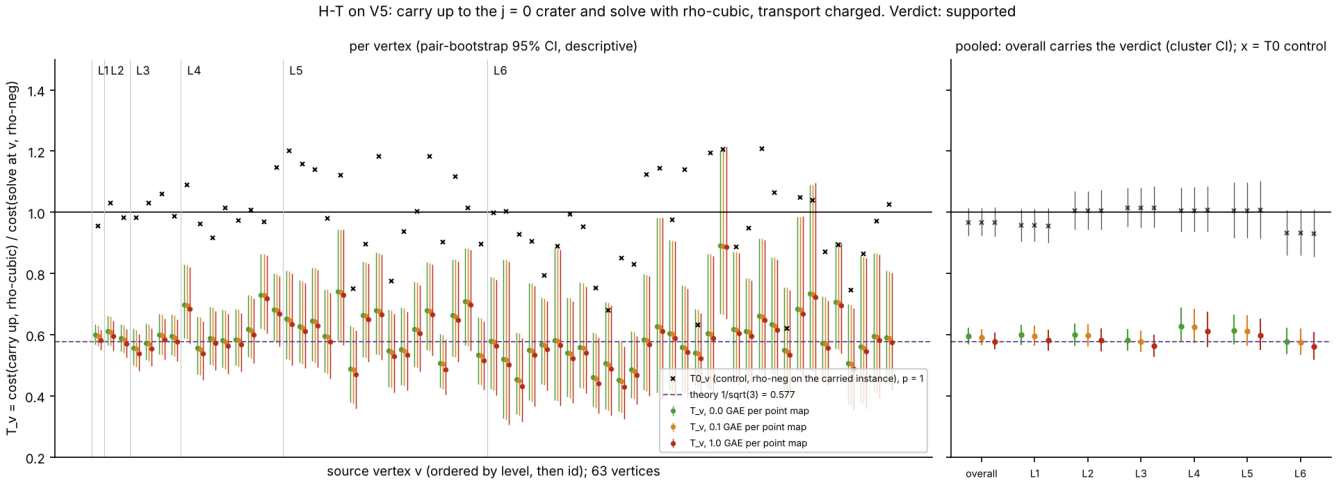


Figure 8: H-T: T_v per source vertex at three prices with the T0 control, and pooled values

fig7_hT: H-T: T_v per source vertex at three prices with the T0 control, and pooled values. Source `analysis/results.json`; SVG beside the PNG.

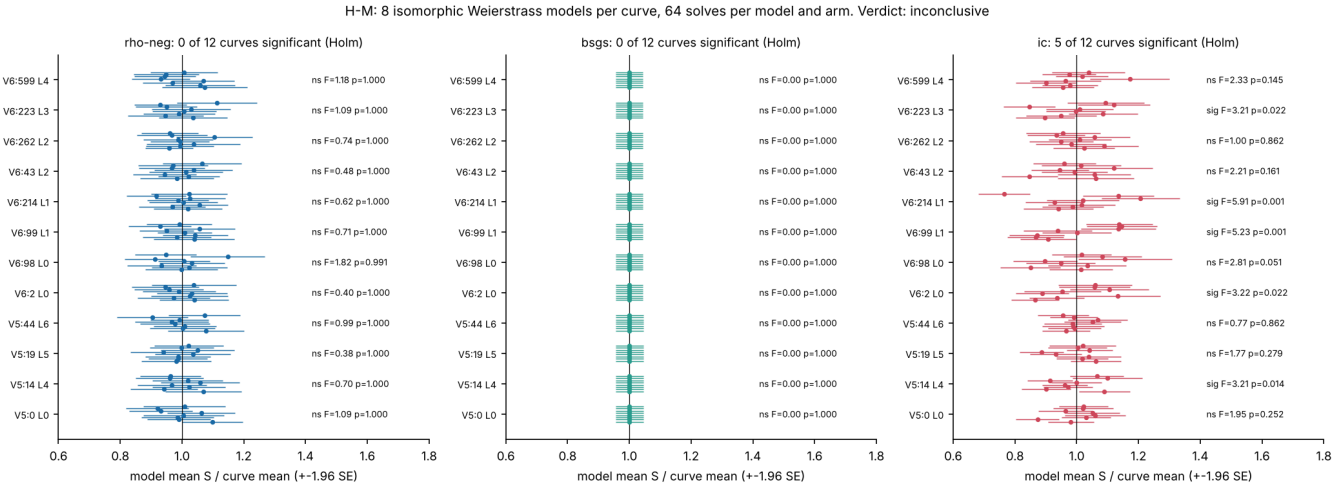


Figure 9: H-M: per-model mean S for rho-neg, bsgs and ic on the 12 chosen curves

fig8_hM: H-M: per-model mean S for rho-neg, bsgs and ic on the 12 chosen curves. Source `analysis/results.json`; SVG beside the PNG.

12. Limits and what is not claimed

- Shared 4-vCPU cloud VM; wall time not admitted (isolation L2 for 84-87% of rows, counted over all rows of the volcano; erratum F4).
- One volcano per question: V5 for the j = 0 results, V6 for the l = 3 replication. Observed, not established across volcanoes.
- V5 levels 1-3 have 1, 2 and 4 curves: descriptive only. The V5 crater is one curve, so Q2a resamples rows, not curves; the slug-cluster variant is a sensitivity.
- The last carry-up edge (level 1 to j = 0) is outside `ca`, done by the study's verified helper; its wall time is not measured and is not part of any result; it is charged as 2 point maps (A3.2).
- **Container restarts and redone sessions.** The driver was restarted 7 times (pauses excluded from the wall budget). Three sessions, `V6-D-L1-c0.r2`, `V6-D-L2-c0.r2`, `V6-N-L1-c0.r2`, were redone after container restarts killed the first attempts; the analysis uses the audited `.r2` sessions, and the three partial first attempts are kept unanalysed as evidence in `runs/orphans/` (V6-D-L1-c0.orphan.tar.xz, V6-D-L2-c0.orphan.tar.xz, V6-N-L1-c0.orphan.tar.xz). Nine busy-host refusals of Tcr sessions (A2.1) are in `progress.jsonl` as `failed` lines and ignored by the inclusion rule; those specs ran later and count once.
- ctl is the negative control and is never pooled; Tcr and M are analysed only in their own sections. The interpretations of A3 are post hoc (written after the data, before the final run).
- H-T compares `rho-cubic` on the carried instance to `rho-neg` at the source, as preregistered; `rho-neg` at the source is the strongest rho available on every one of the 63 (j != 0) source curves, so the denominator is the matched, strongest rho (erratum F3). It is a measurement of a known composition, not a new method. The vertex-weighted pooling (A3.2) weights level 6 most.
- H-M uses 8 models for each of 12 curves and a conservative unpaired permutation; the verdict is on the Monte Carlo boundary between inconclusive (5 of 12, frozen seed) and supported for IC (6 of 12, precise evaluation; erratum F1), so no claim about the cause (x-coordinate dependence of `prime-abscissa`) is made.
- Uncharged counters are priced at 0 / 0.1 / 1 GAE for both arms; counters that `cost.unpriced` does not name are not counted, so both arms stay lower bounds. No cross-method rank is stated where the order does not hold across the band.
- No claim about cryptographic sizes: 26-bit synthetic curves with planted known answers. The transport-then-solve 'signal' cells (section 6) are not a speedup claim.

- This file's analysis plan was frozen before the first Phase B session; the three-restart and A2 operational changes are recorded in `AMENDMENTS.md`.

13. Reproduce

`python3 -I analysis/extract_uncharged.py <scratch>` (one archive at a time), `python3 -I analysis/analyze.py`, `python3 -I analysis/figures.py`, `python3 -I analysis/build_report.py`, `bash analysis/build_pdf.sh`. Input hashes are in `analysis/results.json`.